BUSINESS DATA MANAGEMENT

Lecture07

Supply Elasticity

Hello friends, in the previous video we had discussed the concept of elasticity of demand. In today's video we will try and understand the same concept of elasticity from the point of view of supply. We'll try and understand what are the key variables that impact supply elasticity, how do you go about computing supply elasticity, what are the different types of supply elasticity. We also discuss some of the curves of supply elasticity. And finally, after discussing a couple of case studies, we will also touch upon the concept of economies of scale and the economies of scope. So let's get started with this. So what is

elasticity of supply. Now price elasticity of supply essentially measures how much quantity supplied changes in response to the change in the price. So if you would remember when we discussed the price elasticity of demand, we were trying to measure how much does the demand change in response to the change in price. Here, the difference is that we are looking at it from a supplier's perspective. We are trying to see it from a producer's perspective as to how much of the quantity supplied changes in response to any change in the price.

We regard supply to be elastic if the quantity supplied changes significantly in response to the price change. It can change either more or less, but if the quantity supplied changes significantly, it is regarded that this particular product has an elastic supply as against the supply being inelastic if the response is only marginal or none. Now, given that the whole supply concept is from the point of view of the seller or the supplier, the whole idea of how much can be supplied is heavily dependent on the flexibility of the sellers to change the quantity of supply. Okay, so when the supply becomes more elastic, which is when the quantity changes significantly, that can only happen if the companies or the firms are able to respond very quickly to any change in price point.

They should be able to do it only if they have enough resources. If they are constrained by resources, they will not be able to make that change in the quantity supplied. Similarly, their ability to make some adjustments to the production also is a factor that makes the supply elastic. Supply becomes inelastic when the sellers are constrained. How can they be constrained?

They have a fixed capacity and they are at, let's say, at 95, 98% of their capacity to produce; in that case, they will not be able to change the supply beyond the capacity, so it becomes inelastic. Or, for instance, if in order to make supply available the production will take a lot of time, be able to produce if their production time will take more time; in that case, the supply becomes inelastic. And finally, if the resources for them to be able to procure in order to make the supply possible, if that is difficult to obtain, again in that scenario the supply becomes inelastic. So what are the variables, what are the factors that impact supply elasticity? In a nutshell, if you have to kind of think about some of the key variables that impact supply elasticity, some of the key ones, and this not being exhaustive, is most importantly the period of time or the time horizon. The longer the time it takes,

the longer the time it takes to increase elasticity because the firms can adjust production. In that case, the supply is far more elastic. A car manufacturer can increase production, for example, more easily over a longer period of time than being able to increase the production in a shorter period of time. So that time period has an impact on elasticity. Okay.

Again, as I mentioned earlier, the ability, the availability of capacity, the more the unused capacity, the more the capacity they possess, the more elastic it can be because they are now able to produce within their capacity constraint. For example, if a factory is operating at, let's say 40, 50, 60%, they can quickly raise the output. Okay. So again, availability of fair capacity is a function, is a variable that impacts supply elasticity.

Again, if you step back and think about it, we are all seeing this from the supplier's perspective. We are not seeing it from a consumer perspective here. Third, availability of inputs. If as a supplier, you are able to easily access inputs in terms of raw materials or even labor sometimes availability of labor is not there and that can constrain your ability to supply in that case your elasticity will be impacted so the easier you are able to access raw material labor the other inputs in order to produce that improves your elasticity as an example availability of skilled labor from a services

means that they will be able to quickly expand catering to the demand. The fourth, the nature of the product. The nature of the product per se can have an impact on the level of elasticity. For example, if the goods that you are producing or supplying are perishable or unique goods, they tend to have more inelastic supply. For example, fresh fruits, vegetables, they tend to

rather quickly. Technological capability again helps improve your capacity. Better technology improves your yield and your better yield can scale up production. So automation helps firms increase their output or supply faster. Therefore, those kinds of products or services will be far more elastic.

Government regulation. Now government as an external factor can also have an impact on elasticity. If there are heavy regulations that constrain your ability to produce, this will have an impact on your elasticity and therefore it reduces elasticity. For example, companies within the mining sector, they face licensing restrictions.

So their ability to expand capacity or new companies to come up and help in the production will take far longer gestation period or longer period of time because these companies are regulated. There are a lot of permissions and protocols that they need to follow before they can even start production. So government regulation is also a variable, a key variable that can impact supply elasticity. So in a nutshell, these are six, seven variables that are most critical, most prominent variables that apply to supply elasticity.

But you as a data analyst have to think about from your own company's perspective, from the industry that you are part of when you are looking at your ability to produce, in terms of how quickly can you cater to the market, how can how can you quickly can you respond to the market uh would highly depend upon which product or service you are in, what is the kind of capacity that you are operating in, what are the kind of inputs or resources that you require to quickly scale up, what is your underlying product, how much is the technology contributing to be able to you as an organization becoming more efficient. And are you a regulated organization? If you're a highly regulated organization, then your ability to be elastic will be far lower versus a company or for firms that are not under heavy regulations.

Now that we've understood the fundamental concept of supply elasticity, similar to the way we had discussed the demand elasticity in terms of how do we compute going about measuring demand elasticity, a very similar concept gets applied when you are trying to compute the price elasticity of supply from the supply side. If you remember when we had discussed the price velocity of demand, the computation was around looking at the change in the quantity demanded divided by the percentage change in the price. From a supply perspective, you would again apply the same logic. The percentage change in quantity supplied divided by the percentage change in price. OK, so let's take an example.

Suppose there's an increase in price of milk because of demand or because of some reason by 10%. Okay, and that raises the amount that the dairy farmers can produce 20% more. What would you consider this as in terms of the elasticity of supply? You would look at the change in the quantity supplied. So earlier if you were able to supply 100 gallons, now you're able to supply 120 gallons.

So the change in quantity supplied is 120 minus 100, which is 20. Now, what is the change in the price? The price changes by 10%. So when you look at the change in the quantity supplied, which is 20, divided by the change in price,

which is 10%, you get the answer as 2, which means this is a highly elastic supply. So again, the way we had looked at from a demand perspective,

If you see it from the supply perspective, if the elasticity is more than one, it is regarded as elastic or highly elastic. If it is less than one, it is regarded as inelastic. Similar to the concept of elasticity when we looked at from a demand perspective, let us look at the supply elasticity classification as well from a supply perspective. So what do we get here? We see that the supply elasticity can be looked upon from five perspectives.

Number one is the, okay, let me just again start sharing the screen. For some reason, the screen got disconnected, right? So the supply elasticity classification table suggests that you can look at elasticity from five perspectives. Number one, Elastic supply.

Number two, inelastic supply. Number three, unit elastic supply, which means the change in the quantity supplied by the change in the price is the same, which is at one. Then you have the perfect elastic supply and then we have the perfect inelastic supply. So let's go one by one and understand each of them a little bit more in detail. When you look at elastic supply, what it means is a small change in price leads to a significant or a substantial change in the quantity supplied.

For example, if you look at, you know, clothing manufacturers' ability to increase their production during festive season, they can rapidly increase their production during festive season. That's a good example would suggest that this is an elastic supply. Mind you that also has to be taken into consideration the fact that They are not constrained by any resources, by any capacity constraints. Let's go to the second one, inelastic supply.

And inelastic supply, as I mentioned earlier, means that a change in the price leads to a relatively small change in the quantity supplied. So it's not very easy to shift production because there is a surge in pricing. Look, think from a point of view of agricultural goods. Can you immediately increase the production of agricultural goods?

No. In the short term, that becomes very difficult. That means from the supply side, this is an inelastic product or goods. Third, unit elastic supply. Similar to the way we had discussed unit elastic demand, the percentage of in supply here, the change in supply percentage equals the percentage change in price.

A 10% fall in price leads to a 10% rise in supply. That means it is unit elastic. It is a perfectly elastic kind of a situation. Fourth, perfectly elastic supply. That means the suppliers are willing to supply any quantity at one fixed price, but not at a lower price.

For example, commodity trading comes across as a perfect example in a competitive market scenario where suppliers are willing to supply any kind of quantity at one fixed price. But if you lower the price, it becomes extremely unprofitable for them to supply because they are anyways running on very thin profit margins. Perfectly inelastic supply. That means the supply remains constant. You cannot change the supply regardless of whether the price is 1 or 100.

Look at it from the point of view of, you know, an example of rare paintings being sold by an artist who is no more there in the world. Or for that matter, let's say there is a, you're, you're selling something, you're selling land enough in an island, so the land supply is fixed. Irrespective of how much the price changes, the supply cannot be increased, so that's an example of a perfectly inelastic supply. Let us look at the next section which is around curves, the supply elasticity curves. And let us see it in the context of the previous one that we talked about: elastic supply, inelastic supply, perfectly elastic supply, perfectly inelastic supply, and the unit elastic of supply. Let's look at, for example, in a unit elastic supply, which is when the elasticity is equal to one, your graph on the right-hand side, number C, you will see that any change in the price

leads to a similar change in the quantity that is supplied. So this is the supply, the upward slope supply curve. So any change in the price leads to a similar quantity of change on the supply side. Any change in price leads to a similar quantity of

change in supply. That is unit elastic supply when they both are of equal proportion.

But when you look at graph number D, you see that a change in price leads to a far higher degree of quantity that is available to be supplied. That means your good, your product is elastic. And in the same breath, if you look at a perfectly elastic supply, that means that you are, you are able to supply infinitely high quantity at any price which is above the base price. So that is a scenario where your quantity can move infinitely versus the price, versus a perfectly inelastic supply is where your quantity cannot move. Irrespective of whatever the price point may be, your supply remains the same. We talked about this example in the previous section when you looked at, you know, rare paintings by a deceased artist. So for that matter, any kind of fixed supply situation which cannot be changed irrespective of the change on the price side. So the supply curves will range from being perfectly inelastic to being perfectly elastic.

Elasticity also varies along the supply curve for companies or firms that have limited production capacity. For low levels of quantity supplied, elasticity is high because the firms can easily utilize the spare capacity, as I mentioned. However, when the capacity is to the brink, when you are at a 90, 95, 98% capacity utilization, then making additional supply available becomes very difficult because you are constrained by your capacity. That makes the elasticity from a supply perspective a lot less. Let's look at a couple of cases and

try to understand this whole supply elasticity a little bit more from a real life perspective. So imagine if there is a new invention because of which the ability of the farmers to increase their crop production goes up significantly. The researchers have come out with some kind of a hybrid crop, in this case wheat, and that crop is now made available because at large, you can scale up the supply. Because of this whole technological advancement, the supply curve now, if you visualize the supply curve, which is the upward sloping curve, that moves to the right.

So you're able to kind of increase your ability to make larger quantities available in the market. However, remember that in this case, given the nature of the product, the demand curve remains largely unchanged because the ability of the consumers to buy that particular crop does not get impacted or affected just because now you have a hybrid product which is available to be produced at large quantities. Consequently, when the supply moves up, when there is a greater amount of product that is made available in the market and your demand remains constant, it does not change to that extent, there is an oversupply situation. If you remember, we talked about equilibrium pricing in one of the few videos in the past.

And we said, hey, in that situation, when you have a surplus situation, in that situation the equilibrium price falls as the equilibrium quantity rises. More quantities are made available to the masses, because of which the price, the equilibrium price point, falls. Now we have to understand that the reason why the underlying product commodity is behaving in such a manner is because it is an inelastic product. Okay. So with any significant decrease in prices, the quantity of wheat sold.

Okay. With any significant decrease in prices, the quantity of wheat sold only increases to some extent. It does not increase significantly. Hence, the total revenue, from a revenue standpoint, when you look at it in terms of a firm that now has a larger number of products to sell because of technology advancements, and the demand remains the same, what happens to the price point is that the price point goes down. Therefore, the total revenue goes down simply because the quantity sold does not increase

in the same proportion to which the decrease or the decline in the pricing. This explains the paradox of public policy that aims to help farmers. It's a paradox because it aims to help farmers by paying them not to plant crops, by reducing the supply. So when you constrain supply, that leads to increase in prices. It leads to increase in revenues.

So that's why good news for the farming industry may sometimes be regarded as bad news for the underlying farmers who are the suppliers or the sellers. Let's take another interesting case study that happened, a real-time case study that happened a few decades back in the 1970s in OPEC, which is the Organization of Petroleum Exporting Countries. This entire OPEC, they reduced the amount of oil, they reduced the amount of supply of oil just to increase the prices and their incomes. As we know, the product per se is inelastic simply because the consumers will want to consume oil. even if the prices increase.

So what happens in the short term? In the short term the supply and demand for oil as we know is relatively inelastic. Therefore supply is inelastic as we know because the oil reserves and extraction capacity cannot be changed overnight. You cannot change, you cannot increase capacity of oil just because there is a shortage of oil, right?

And the prices move up, it's difficult. Therefore a shift in the supply curve leads to a massive increase in prices. Now this shift is the decline in the supply. So when demand remains constant, and your supply goes down, that's a situation of shortage.

In a situation of shortage, the prices move up and that leads to higher revenues for the players in OPEC in the short term. However, in the long term, what happens? In the long term, what happened was like over a period of time, producers that were those outside the OPEC, countries, okay? They responded to increasing oil exploration.

They built their capacity. They built their extraction capacity while consumers also very slowly responded to conserving energy and replacing the inefficient old vehicles, cars and other vehicles with newer, more efficient ones. So what are we saying? We are saying that because in the long run supply and demand curves are much more elastic compared to the short term.

So the time horizon concept again comes in over here because of capacity constraint, to immediately jump in and increase their production, they are

constrained because it is not easy to set up extraction facilities and oil exploration facilities. Over the long term, supply and demand curves are regarded as far more elastic compared to the short term. The same reduction in supply that OPEC eventually, this whole reduction that the organized reduction by OPEC countries that happened, they eventually caused a much smaller increase in the price of oil, causing prices to steadily decline. as time went by in the long term the prices declined steadily but in the short term they did increase and they did fill the coffers of these companies so it's important to understand this case study from the point of view of capacity from the point of view of the time horizon

Finally, having understood the whole supply-side scenario and supply-side elasticity, it's important to understand from the supply perspective the concept of economies of scale and the economies of scope. And this you have to see from the point of view of, again, a supplier's ability to supply profitably out in the market. And how can the suppliers be more profitable through the concept of economies of scale and economies of scope? Now, economies of scale primarily refers from a supply-side perspective, which means basically the firms or the companies

are able to produce or produce at large scales or large quantities so when a producer is able to cater to the market larger quantities it's important to understand that their average cost of producing those large quantities the average cost per unit of producing those large quantities goes down that is simply because the fixed costs which are fixed which do not change and we will discuss the concept of fixed and variable costs in far more detail in the in the upcoming finance and accounts module but at a high level it's important for you to understand that the average cost of production falls as the number of units increased from a supplier side when they produce more the average cost per unit goes down when the average cost goes down and the price remains the same or similar the profitability improves for the suppliers also because you are able to now cater to a much wider market or much larger supplies and your cost goes

down, your ability to also lower the prices is far more and that essentially gives you as a supplier a competitive advantage. This causes the supply curve to shift to the right.

The supply curves shift to the right with larger quantities And then the price point potentially can also improve in terms of from a customer perspective. But most importantly, the profit margins of the company improves because of economies of scale. That's a reason, one of the reasons why companies offer bulk discounts when you buy in large quantities. Okay.

Now, what is the construct of economies of scope? But think about firms and organizations that are producing a wide variety of or a variety of products or similar products. When they have the ability to share resources in order to produce a wide variety of products, that leads to economies of scope. They're able to now bunch together multiple products. And because of being able to bunch together multiple products, they're able to reduce the production costs.

How are they able to reduce the production costs? They're able to reduce the production costs as they are able to save the costs by sharing of resources. Now the same set of individuals who are catering to one product in your company, if they're able to cater to three products of your company, now you are able to allocate the resource into three products and not just one product. So that is how sharing of resources across products leads to savings of costs and it reduces the production costs.

What happens when you do that? Your ability to be more efficient as a production unit becomes more. And finally, you are able to increase your supply across not just one product category, but across a wide range of product categories, which ultimately I would say dovetails into economies of scale where you're able to provide the larger supplies at better cost, your profitability improves, and also your ability to make the pricing more attractive also improves, leading to competitive advantage. So I think both these concepts are separate, economies

of scale and economies of scope, but they also dovetail or converge with each other in the end. So with that,

I would say that we have finally come to the end of this particular section on demand and supply, where we discussed a wide variety of concepts, including demand curves, supply curves, elasticity, economies of scale and scope, I hope all of you got an opportunity to not just learn, but also thoroughly enjoyed learning economics. This is obviously a very basic and fundamental level of learning. As we move forward in this program, in future programs, we will go far more deeply into both micro and macro economics. So thank you very much and I'll see you in the future videos.